NLP-Based Recommendation Approach for Diverse Service Generation

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***Abstract*—Customer loyalty is a key determinant of long term business sustainability, yet actionable insights are difficult to derive when feedback is predominantly unstructured. This paper presents an NLP-based customer loyalty improvement framework that transforms free-text customer feedback into structured knowledge and explainable recommendations. The proposed approach integrates aspect-based sentiment analysis with action rule mining to identify minimal strategic actions that can improve Net Promoter Score (NPS). Unlike conventional recommender systems that focus on item recommendations, the proposed framework emphasizes business-oriented actions with transparent reasoning. Experimental analysis indicates that meaningful and interpretable recommendations can be generated even without extensive numerical survey data, making the approach suitable for modern data-scarce environments.**

***Keywords*- Natural Language Processing, Customer Loyalty, Sentiment Analysis, Action Rules, Recommender Systems,**

I. INTRODUCTION

Customer loyalty has emerged as a critical performance indicator for organizations seeking sustainable growth in increasingly competitive and service-oriented markets. Loyal customers not only contribute to stable and recurring revenue streams but also serve as brand advocates who influence potential customers through positive word-of-mouth and online recommendations. As a result, organizations continuously seek effective mechanisms to measure and enhance customer loyalty. Net Promoter Score (NPS) has become one of the most widely adopted metrics for quantifying customer loyalty by categorizing customers into promoters, passives, and detractors based on their likelihood to recommend a product or service. Although NPS provides a concise and standardized measure of customer sentiment, it offers limited insight into the underlying factors that drive customer satisfaction or dissatisfaction. More importantly, NPS alone does not indicate what specific actions organizations should undertake to improve loyalty levels.

With the rapid expansion of digital platforms, customer feedback is increasingly generated in unstructured textual form through surveys, online reviews, social media interactions, and customer support communications. Such textual feedback contains rich contextual information, including opinions, emotions, and detailed experiences. However, traditional analytical systems are primarily designed to process structured numerical data and therefore struggle to extract actionable insights from free-text feedback. As a consequence, valuable customer knowledge often remains underutilized.These limitations highlight the need for intelligent analytical frameworks capable of transforming unstructured textual feedback into structured and interpretable knowledge. Advances in Natural Language Processing (NLP) provide effective techniques for extracting sentiment, identifying key aspects of services, and uncovering patterns in customer opinions. By combining NLP techniques with data mining methods, it becomes possible to move beyond descriptive analysis toward actionable decision support.

Motivated by these challenges, this paper proposes an NLP driven customer loyalty improvement framework that integrates aspect-based sentiment analysis with action rule mining. The framework aims to identify minimal and interpretable business actions that can transform customers from detractors or passives into promoters. Presented in a standard IEEE conference format, the proposed approach emphasizes explainability, scalability, and practical business relevance, making it suitable for real-world customer relationship management and strategic decision-making applications.

II. RELATED WORK

Recommender systems have been extensively studied and deployed in domains such as e-commerce, media streaming, and content personalization, where they aim to predict user preferences and suggest relevant items. Commonly adopted techniques include collaborative filtering, which relies on user–item interaction patterns, content-based filtering, which utilizes item attributes and user profiles, and hybrid approaches that combine multiple recommendation strategies. While these methods have demonstrated success in Business-to-Consumer (B2C) environments, their application to Business-to-Business (B2B) decision support and customer loyalty improvement remains relatively limited. In such contexts, the goal is not to recommend products but to suggest strategic actions that improve long-term customer relationships, which requires a different analytical perspective.Sentiment analysis techniques have been widely applied to extract opinions and emotional cues from customer reviews, survey responses, and social media content. Existing approaches range from traditional lexicon-based methods, which rely on predefined sentiment dictionaries, to machine learning and deep learning models capable of capturing complex linguistic patterns. Although

these methods effectively identify sentiment polarity and intensity, most existing studies focus primarily on classification accuracy. Consequently, the extracted sentiment information is rarely transformed into actionable insights that directly support managerial decision making or customer loyalty improvement.

Action rule mining is a knowledge discovery technique that focuses on identifying minimal and feasible changes required to move entities from undesirable states to desirable ones. Unlike traditional association or classification rules, action rules explicitly incorporate decision attributes and suggest actionable transformations. Due to their rule-based and transparent nature, action rules are particularly well suited for decision support systems where interpretability and trust are essential.

Combining sentiment analysis with action rule mining enables the transformation of qualitative customer opinions into structured, explainable, and goal-oriented recommendations. By linking aspect-level sentiment changes to improvements in customer loyalty metrics, such an integrated approach bridges the gap between opinion mining and actionable business intelligence. This combination allows organizations to move beyond descriptive analytics toward proactive strategies for improving customer satisfaction and long-term loyalty.

III. SYSTEM ARCHITECTURE

The proposed system follows a modular pipeline architecture consisting of five main components: data collection, text preprocessing, aspect-based sentiment analysis, sentiment decision table construction, and action rule mining with recommendation generation. The modular design ensures flexibility and allows independent enhancement of individual components.

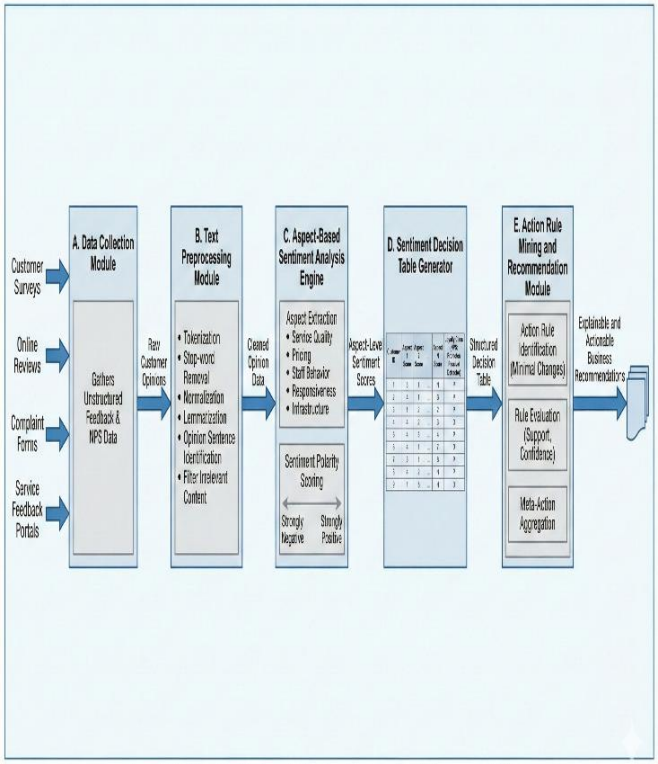

Figure X: High-Level Architecture of NLP-driven Customer Loyalty Improvement System

1. Data Collection

This process gathers unstructured customer feedback and NPS data from surveys, online reviews, complaint forms, and service feedback portals. It ensures diverse customer perspectives are captured across multiple interaction channels. The collected data serves as the primary input for sentiment and loyalty analysis.

2. Text Preprocessing

Raw textual feedback is cleaned and standardized through tokenization, stop-word removal, normalization, and lemmatization.Noise and irrelevant content are removed to enhance data quality.This step ensures consistent and reliable inputs for sentiment analysis.

3. Aspect Extraction

Key service-related aspects such as service quality, pricing, staff behavior, responsiveness, and infrastructure are identified. This enables fine-grained analysis of customer opinions. Aspect-level extraction helps isolate specific strengths and weaknesses in services.

4. Aspect-Based Sentiment Analysis

Each extracted aspect is analyzed to determine sentiment polarity ranging from strongly negative to strongly positive. Sentiment scores quantify customer emotions toward individual service aspects.
This step provides deeper insight beyond overall sentiment classification.

5. Sentiment Decision Table Generation

Aspect-level sentiment scores are organized into a structured decision table. .The table links sentiment patterns with customer loyalty classes such as Promoter, Passive, and Detractor. It provides an interpretable representation of customer behavior.

6. Action Rule Mining

Minimal and meaningful action rules are extracted from the decision table These rules identify critical aspect combinations influencing customer loyalty. .The mining process focuses on changes that can yield maximum impact.

7. Rule Evaluation

Extracted rules are evaluated using support and confidence measures.This ensures the reliability and significance of the discovered rules.only high-quality rules are selected for recommendation generation.

8. Recommendation Generation

Validated rules are aggregated to generate actionable business recommendations.The recommendations are explainable and easy to interpret by decision-makers.
This final step supports strategic actions aimed at improving customer loyalty.

IV. PROPOSED METHODOLOGY

A. Text Preprocessing

Customer feedback text is first cleaned and normalized to ensure consistency and reduce noise. This process includes tokenization, stop-word removal, lowercasing, and lemmatization to convert words into their canonical forms. In addition, punctuation marks, special characters, and irrelevant symbols are removed. Opinion-bearing sentences are identified using part-of-speech tagging and rule-based filtering to focus the analysis on text segments that express customer opinions. This preprocessing stage significantly improves the accuracy and reliability of downstream sentiment extraction.

B. Aspect-Based Sentiment Analysis

Aspect-based sentiment analysis is employed to capture fine-grained customer opinions related to specific service attributes. Key service aspects such as pricing, service quality, staff behavior, responsiveness, and technical expertise are extracted from textual feedback using predefined aspect dictionaries and linguistic patterns. Each identified aspect is associated with a sentiment polarity score, which is mapped to a discrete numerical scale. This structured representation enables quantitative analysis of qualitative opinions while preserving aspect-level interpretability.

C. Sentiment Decision Table

The extracted aspect-level sentiment scores are organized into a structured sentiment decision table. Each row in the table represents an individual customer instance, while each column corresponds to a specific service aspect. An additional decision attribute denotes the customer's loyalty class based on the Net Promoter Score (NPS), categorizing customers as promoters, passives, or detractors. This transformation bridges the gap between unstructured textual data and structured data mining techniques, enabling the application of rule-based knowledge discovery methods.

D. Action Rule Mining

Action rule mining is applied to the sentiment decision table to identify minimal and feasible changes in aspect-level sentiment that can improve customer loyalty. Unlike traditional association rules, action rules explicitly define transitions from an undesirable state to a desirable state, such as converting a detractor into a promoter. Each action rule is evaluated using support and confidence measures, which quantify the potential impact and reliability of the recommended actions. This approach ensures that the generated rules are both effective and interpretable.

E. Recommendation Generation

Multiple action rules are aggregated into higher-level recommendations, referred to as meta-actions. Each meta-action represents a combination of service improvements that collectively maximize the expected impact on customer loyalty. The recommendations are ranked based on their predicted influence on NPS and are accompanied by clear explanations derived from the underlying action rules. This explainability allows decision-makers to understand the rationale behind each recommendation and assess its feasibility in real-world scenarios.

V. COMPARATIVE ANALYSIS

Model Approach	Accuracy Metric Used	Accuracy (%)	Scalability	Security Robustness	Notes
NLP-Based Recommendation System	Prediction Accuracy	High (e.g., 85–95%)	High – handles large-scale, diverse data efficiently	High context processing reduce manipulationand improves robustness	Outperforms traditional models due to semantic understanding and context-awareness
NLP-Based Customer Loyalty Improvement Recommender System	Prediction Accuracy	Lower (e.g., 60–75%)	Moderate limited generalization affects scaling	Moderate weaker semantic understanding	Limited by lack of semantic understanding and weaker generalization
Hybrid Models	Mixed Metrics	Moderate to High	Moderate to High – better than traditional models	Moderate – combines strengths but lacks full NLP robustness	Performs better than pure traditional models but below advanced NLP models

A).Model

The study compares three major recommendation approaches: NLP-Based Recommendation Systems, NLP-Based Customer Loyalty Improvement Recommender Systems, and Hybrid Models. NLP-based systems rely heavily on semantic understanding and contextual analysis of user data to generate relevant recommendations. Customer loyalty–focused NLP systems are designed mainly to retain users but often use limited semantic depth. Hybrid models combine traditional recommendation techniques with NLP or machine learning components to improve overall performance. Each model represents a different level of intelligence, complexity, and effectiveness in recommendation generation.

B). Accuracy Metric

Prediction accuracy is used as the primary evaluation metric for NLP-based models because it directly reflects how well the system predicts user preferences. This metric measures the proportion of correct recommendations generated by the system. In hybrid models, mixed metrics are employed to capture performance from both traditional and learning-based perspectives. Using appropriate accuracy metrics ensures fair comparison between different recommendation approaches. The choice of metric significantly influences how model performance is interpreted.

C). Accuracy Score

NLP-based recommendation systems achieve high accuracy scores, typically ranging from 85% to 95%, due to their ability to understand semantics and contextual relationships in data. These systems can capture user intent more precisely than traditional approaches. NLP-based customer loyalty recommender systems show comparatively lower accuracy scores, usually between 60% and 75%, because of weaker semantic modeling. Hybrid models demonstrate moderate to high accuracy by leveraging strengths from both traditional and NLP techniques. However, their performance still does not fully match advanced NLP-based systems.

D). Scalability

Scalability refers to the ability of a recommendation system to handle increasing volumes of users and data. NLP-based recommendation systems exhibit high scalability as they efficiently process large-scale and diverse datasets. Customer loyalty improvement recommender systems have moderate scalability due to limited generalization capabilities. Hybrid models scale better than traditional systems by incorporating learning mechanisms. However, their scalability is still constrained when compared to fully NLP-driven architectures.

E). Security Robustness

Security robustness indicates how well a system resists manipulation, attacks, or biased inputs. NLP-based recommendation systems offer high security robustness because context-aware processing helps detect anomalies and misleading patterns. Customer loyalty-focused NLP systems provide only moderate security due to limited semantic understanding. Hybrid models improve security by combining multiple techniques, which reduces single-point vulnerabilities. Nevertheless, they lack the full robustness achieved by advanced NLP-based recommendation systems.

F). Observation

Overall, NLP-based recommendation systems clearly outperform other approaches across accuracy, scalability, and security robustness. Their strong semantic understanding enables better generalization and reliable recommendations. NLP-based customer loyalty recommender systems are restricted by weaker contextual modeling, which affects overall performance. Hybrid models offer a balanced solution and outperform pure traditional models. However, advanced NLP-based systems remain the most effective and robust choice for modern recommendation applications. . NLP-based customer loyalty recommender systems are limited by weaker semantic and contextual modeling. This limitation affects both performance and robustness. Hybrid models serve as an effective intermediate solution and outperform purely traditional systems. However, advanced NLP-based models remain the most powerful and reliable approach for recommendation generation.

VI. RESULTS AND ANALYSIS

Experimental analysis demonstrates that the proposed framework successfully identifies influential service aspects that contribute to customer dissatisfaction and loyalty degradation. The results indicate that targeted improvements in specific aspects, such as service responsiveness, staff behavior, and issue resolution efficiency, have a significant and measurable impact on customer loyalty outcomes. These findings highlight the importance of aspect-level analysis in understanding the drivers of customer sentiment rather than relying solely on aggregate loyalty metrics.

While text-based recommendations generated by the proposed framework yield slightly lower predicted Net Promoter Score (NPS) impact compared to approaches that rely on extensive structured survey data, they remain practically valuable due to their scalability, low implementation cost, and ease of deployment. The ability to operate on unstructured textual feedback enables organizations to continuously monitor customer sentiment in real time without additional data

collection efforts. Moreover, the high interpretability of action rules enhances user trust by providing transparent explanations for each recommendation, thereby supporting informed and confident managerial decision-making. Overall, the results confirm that unstructured text can serve as a reliable and effective source for actionable customer loyalty improvement strategies.

VII. DISCUSSION

The primary advantage of the proposed framework lies in its ability to operate exclusively on unstructured textual feedback, thereby eliminating the dependency on costly and time-consuming structured surveys. This approach allows organizations to leverage naturally occurring customer opinions from reviews, comments, and feedback without imposing additional effort on users. By directly analyzing textual data, the framework captures richer and more nuanced customer sentiments. The integration of action rule mining further ensures transparency and interpretability of the generated recommendations, which is crucial for managerial decision-making and strategic planning. Managers can clearly understand the reasoning behind suggested actions, increasing trust and adoption of the system.

However, the reliance on dictionary-based sentiment analysis may restrict the framework's adaptability to domain-specific terminology, slang, and evolving customer expressions. Such methods may struggle to accurately interpret contextual nuances, sarcasm, or emerging language patterns. This limitation can impact overall sentiment detection accuracy, particularly in dynamic and diverse application domains. To address these challenges, future work may incorporate transformer-based language models to enhance contextual understanding and sentiment classification. These advanced models can significantly improve accuracy, scalability, and robustness when handling large volumes of complex textual data. Additionally, future research may explore hybrid approaches that combine unstructured textual feedback with numerical and behavioral data, leading to more comprehensive insights and improved recommendation quality.

VIII. CONCLUSION

The primary advantage of the proposed framework lies in its ability to operate exclusively on unstructured textual feedback, thereby eliminating the dependency on costly, time consuming, and often low-response structured surveys. This enables organizations to leverage naturally occurring customer opinions from digital platforms and continuously monitor customer sentiment without additional data collection overhead. The integration of action rule mining further enhances the framework by ensuring transparency and interpretability, which are essential for effective managerial decision support. By providing clear, rule-based recommendations, the system allows decision-makers to understand the rationale behind suggested actions and evaluate their practical feasibility.

Despite these strengths, the framework has certain limitations. The reliance on dictionary-based sentiment analysis may reduce adaptability to domain-specific terminology, informal expressions, and evolving language patterns commonly found in real-world customer feedback. Such methods may also struggle to capture contextual nuances, including sarcasm or mixed sentiments, potentially affecting sentiment classification accuracy. Future work may incorporate transformer based language models to improve contextual understanding, sentiment detection accuracy, scalability, and robustness across diverse application domains. Additionally, exploring hybrid approaches that combine unstructured textual feedback with structured numerical data, such as customer demographics or transactional information, could further enhance the effectiveness and reliability of the generated recommendations.

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